

My title*

My subtitle if needed

Shrey Sati

September 27, 2026

First sentence. Second sentence. Third sentence. Fourth sentence.

1 Introduction

Overview paragraph

Estimand paragraph

Results paragraph

Why it matters paragraph

Telegraphing paragraph: The remainder of this paper is structured as follows. Section [2](#)....

2 Data

2.1 Overview

We use the statistical programming language R (R Core Team 2026).... Our data (Toronto Public Health 2026).... Following Alexander (2023), we consider...

Overview text

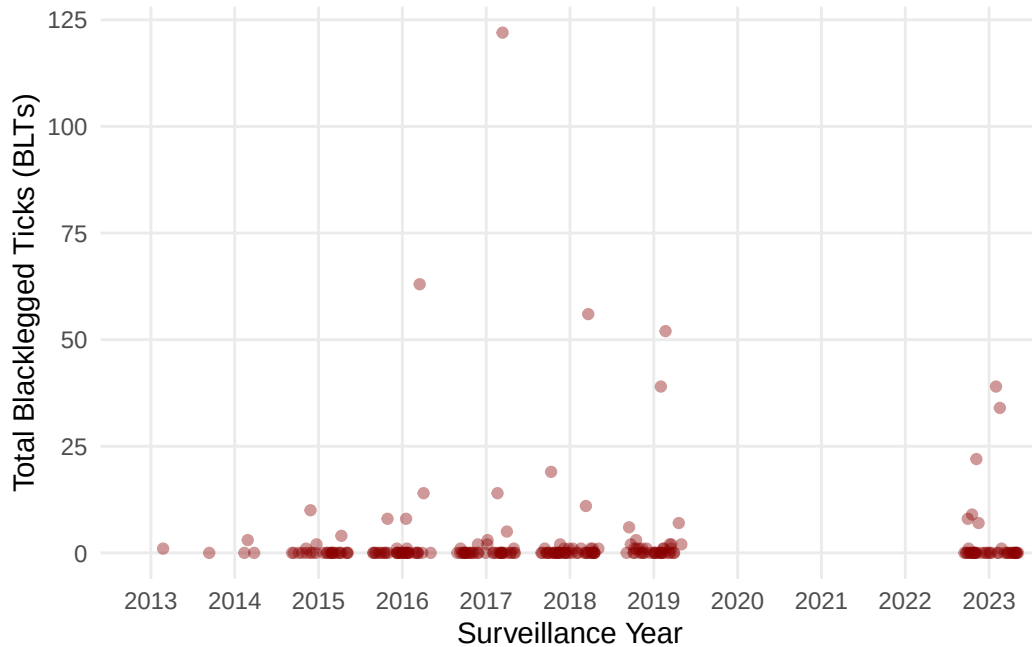


Figure 1: Distribution of total blacklegged ticks collected per park across surveillance years in Toronto.

2.2 Measurement

Some paragraphs about how we go from a phenomena in the world to an entry in the dataset.

```
# Aggregate unique park locations with their total ticks and coordinates
park_summary <- analysis_data |>
  group_by(park_locations, latitude, longitude) |>
  summarize(
    total_blts_all_years = sum(total_blts, na.rm = TRUE),
    .groups = "drop"
  )

# Scatter plot using geographic coordinates
ggplot(park_summary, aes(x = longitude, y = latitude)) +
  geom_point(
    aes(size = total_blts_all_years, color = total_blts_all_years),
    alpha = 0.75
  ) +
```

*Code and data are available at: https://github.com/UpCatch/Toronto_Tick_Surveillance_Data.

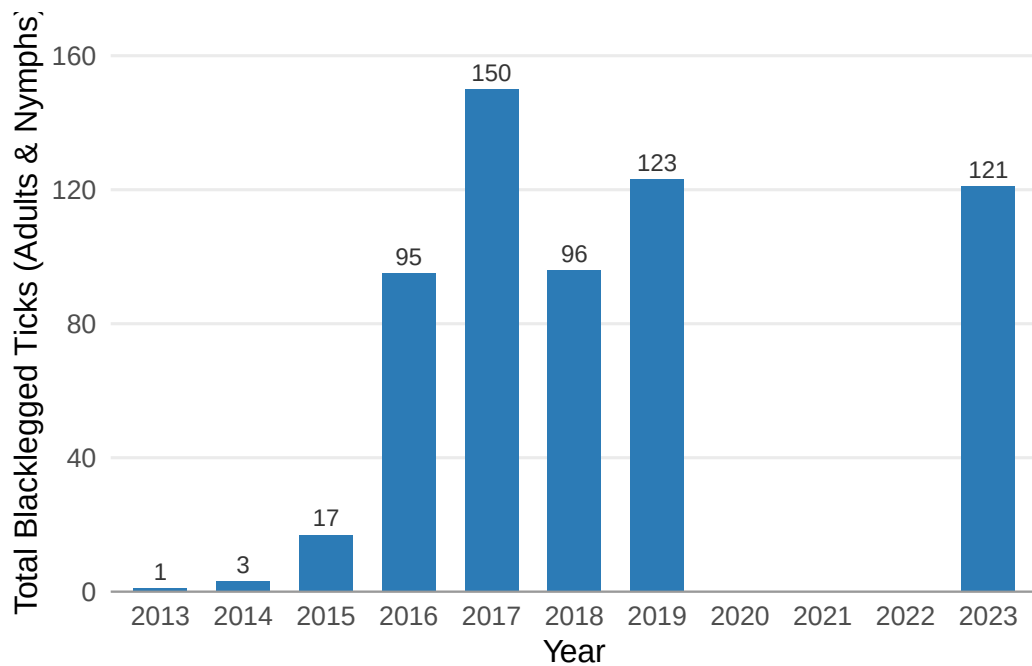


Figure 2: Total adult and nymph blacklegged ticks collected per year across all surveyed parks in Toronto.

```
scale_color_viridis_c(
  option = "plasma",
  trans = "pseudo_log",
  name = "Total BLTs"
) +
scale_size_continuous(
  range = c(2, 6),
  name = "Total BLTs"
) +
guides(
  color = guide_legend(),
  size = guide_legend()
) +
coord_quickmap(
  xlim = c(-79.62, -79.14),
  ylim = c(43.59, 43.83)
) +
labs(
  x = "Longitude (°W)",
```

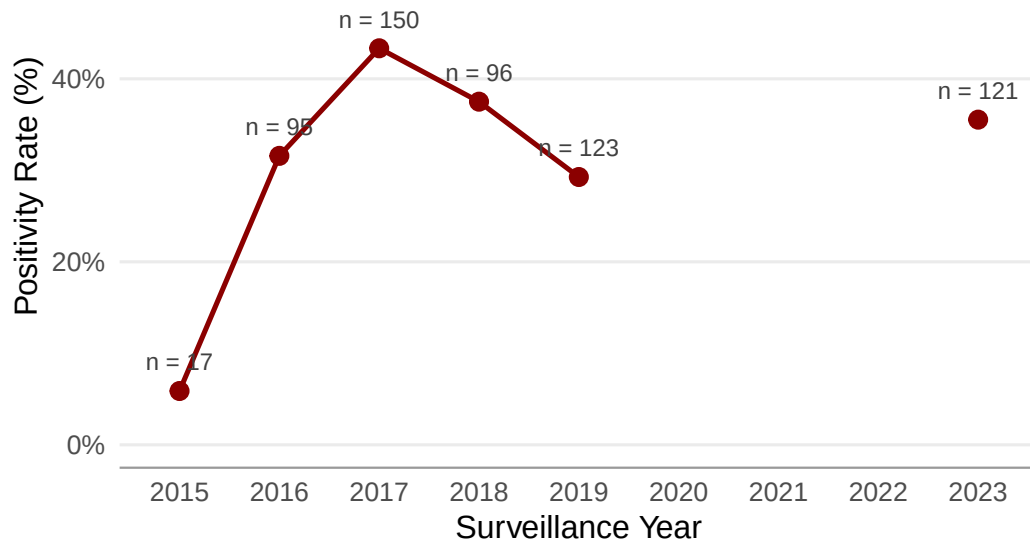


Figure 3: Annual blacklegged tick positivity rate across Toronto parks (2015–2023), with the number of positive ticks annotated.

```

y = "Latitude (°N)",
title = "Blacklegged Tick Surveillance Locations in Toronto",
subtitle = "Point size and color indicate cumulative ticks collected across all surveill
) +
theme_minimal(base_size = 12) +
theme(
  panel.grid.minor = element_blank(),
  panel.background = element_rect(fill = "#fcfcfc", color = NA),
  plot.title = element_text(face = "bold", size = 13),
  plot.subtitle = element_text(color = "grey35", margin = margin(b = 10)),
  legend.position = "right"
)

```

Blacklegged Tick Surveillance Locations in Toronto

Point size and color indicate cumulative ticks collected across all

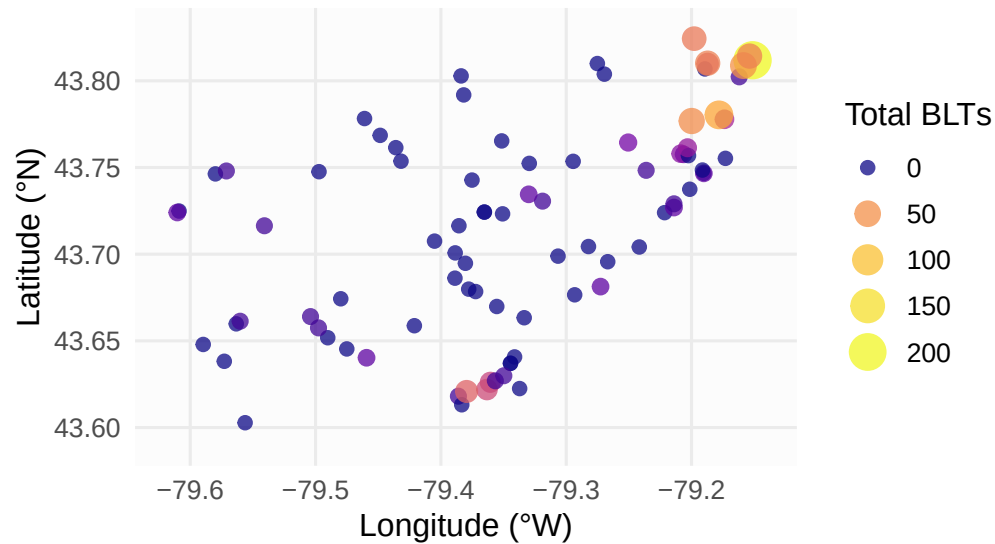


Figure 4: Geographic distribution of blacklegged tick surveillance locations across Toronto parks.

References

- Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman; Hall/CRC. <https://tellingstorieswithdata.com/>.
- R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://doi.org/10.32614/R.manuals>.
- Toronto Public Health. 2026. *Blacklegged Tick Surveillance*. <https://open.toronto.ca/dataset/blacklegged-tick-surveillance/>.